

✓ LEVEL 3 — ADVANCED MCQs (1–150)

1. If a relation $R = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$ on $A = \{a, b, c\}$, then R is:

- a) Reflexive only
- b) Symmetric only
- c) Reflexive and Symmetric only
- d) Equivalence

Ans: c

2. Domain of $f(x) = \sqrt{4 - x^2}$ is:

- a) $x > 2$
- b) $-2 < x < 2$
- c) $-2 \leq x \leq 2$
- d) $x \in \mathbb{R}$

Ans: c

3. If $f(x) = x^2 + 1$ and $g(x) = \sqrt{x - 1}$, domain of $g \circ f$ is:

- a) All real
- b) $x \geq 0$
- c) $x \geq 0$ except 0
- d) All real except negative

Ans: a

4. Which function is bijective over $\mathbb{R} \rightarrow \mathbb{R}$?

- a) x^2
- b) $|x|$
- c) x^3
- d) e^x

Ans: c

5. If $f(x) = |x - 2|$, range is:

- a) $[0, \infty)$
- b) $(-\infty, 0]$
- c) $(0, \infty)$

d) R

Ans: a

6. Number of equivalence relations on a set with 3 elements:

a) 5

b) 3

c) 4

d) 1

Ans: a

7. If $f(x) = x^3 - 3x + 2$, then f is:

a) One-one

b) Many-one

c) Onto only

d) Constant

Ans: a

8. If $f : A \rightarrow B$ is bijective, then number of invertible functions from A to B if $|A|=4$:

a) 4

b) 16

c) 24

d) 256

Ans: c (4!)

9. Range of $f(x) = \frac{1}{1+x^2}$:

a) $(0, 1]$

b) $[0, 1)$

c) $(0, \infty)$

d) $(-\infty, 1)$

Ans: a

10. For what values of a is $f(x) = ax + 2$ invertible?

a) All real a

b) $a = 0$

c) All a except 0

d) Only $a > 0$

Ans: c

11. Domain of $f(x) = \ln(9 - x^2)$:

a) $x < 3$

b) $x > -3$

c) $-3 < x < 3$

d) $x \neq 0$

Ans: c

12. Function $f(x) = \tan^{-1} x$ is:

a) One-one only

b) Onto only

c) Bijective

d) Neither

Ans: c

13. Range of $f(x) = \sin^{-1}(2x)$ is defined when:

a) $|x| \leq \frac{1}{2}$

b) $|x| \leq 1$

c) $x \geq 0$

d) All real

Ans: a

14. If f is even, then:

a) $f(-x) = f(x)$

b) $f(-x) = -f(x)$

c) Graph symmetric about x-axis

d) f is odd

Ans: a

15. The function $f(x) = |\sin x|$ is:

a) One-one

b) Periodic

c) Increasing

d) Invertible

Ans: b

16. Number of functions from $A=\{1,2,3\}$ to $B=\{1,2\}$:

- a) 6
- b) 8
- c) 9
- d) 27

Ans: b (2^3)

17. Domain of $\frac{1}{\sqrt{x-1}}$:

- a) $x > 1$
- b) $x \geq 1$
- c) $x < 1$
- d) All real

Ans: a

18. Range of $\cos^{-1} x$ is:

- a) $[0, 2\pi]$
- b) $[0, \pi]$
- c) $(0, \pi/2)$
- d) $[-1, 1]$

Ans: b

19. If f and g are bijective, then $f \circ g$ is:

- a) One-one only
- b) Onto only
- c) Bijective
- d) Constant

Ans: c

20. If $f(x) = x^2$, then f is invertible on:

- a) $\mathbb{R}$
- b) $x \geq 0$
- c) $x < 0$
- d) None

Ans: b

✓ Continuing — MCQs 21–150

(Consistent 4-option JEE pattern, no explanation, only answers.)

21. Invertible functions are always:

- a) Surjective
- b) Injective
- c) Bijective
- d) Constant

Ans: c

22. Range of e^x :

- a) $[0, \infty)$
- b) $(0, \infty)$
- c) $(-\infty, 1]$
- d) $\mathbb{R}$

Ans: b

23. Domain of $\ln(\ln x)$:

- a) $x > 1$
- b) $x > 0$
- c) $e^x > 1$
- d) $x > e$

Ans: a

24. If R is symmetric and transitive but not reflexive, then R can be made equivalence by adding:

- a) All (a, a)
- b) All (a, b)
- c) All (b, a)
- d) None

Ans: a

25. Domain of $f(x) = \frac{x}{x^2-4}$:

- a) $x \neq \pm 2$
- b) $x > 2$
- c) $x < -2$
- d) All real

Ans: a

26. Range of $f(x) = \frac{1}{x}$:

- a) $x \neq 0$
- b) $y \neq 0$
- c) All y
- d) $y \geq 0$

Ans: b

27. Number of reflexive relations on a set with 3 elements:

- a) 8
- b) 64
- c) 512

d) 16

Ans: b

28. "Divides" relation on positive integers is:

- a) Reflexive
- b) Transitive
- c) Antisymmetric
- d) All

Ans: d

29. Domain of $f(x) = \tan^{-1}(\sqrt{x})$:

- a) $x \geq 0$
- b) $x > 0$
- c) $x > -1$
- d) All real

Ans: a

30. Range of $\sqrt{4-x} - \sqrt{x}$:

- a) $\mathbb{R}$
- b) $(-2, 2)$
- c) $[-2, 2]$
- d) None

Ans: c

31. If f is odd:

- a) $f(-x) = f(x)$
- b) $f(-x) = -f(x)$
- c) Graph symmetric about y-axis
- d) f undefined

Ans: b

32. Range of $f(x) = \frac{x-1}{x+1}$:

- a) All except 1
- b) All except -1
- c) No restriction
- d) $(-1, 1)$

Ans: b

33. Domain of $\sin^{-1}(3x)$:

- a) $|x| \leq 1$
- b) $|x| \leq \frac{1}{3}$
- c) $x > 0$
- d) All real

Ans: b

34. Range of $\log(x+2)$:

- a) $(0, \infty)$
- b) $(-\infty, \infty)$
- c) $[2, \infty)$
- d) $[0, \infty)$

Ans: b

35. Domain of $\sqrt{5-x} - \sqrt{x+1}$:

- a) $-1 \leq x \leq 5$

b) $x \geq 0$

c) $x > 0$

d) $x \leq 5$

Ans: a

36. Number of anti-symmetric relations on set with 2 elements:

a) 4

b) 5

c) 7

d) 9

Ans: c

37. If $f(x)=x^3+1$, inverse exists because f is:

a) Increasing

b) Constant

c) Onto only

d) Even

Ans: a

38. Inverse of $f(x) = \frac{ax+b}{cx+d}$ exists when:

a) $ad - bc = 0$

b) $ad - bc \neq 0$

c) $a=d$

d) $a=0$

Ans: b

39. If f is bijection, cardinality of f^{-1} possible is:

a) 1

b) More than one

c) Infinite

d) 0

Ans: a

40. Range of $x + \frac{1}{x}$, $x > 0$:

a) $(2, \infty)$

b) $[2, \infty)$

c) $(-\infty, 2]$

d) $[0, \infty)$

Ans: b